

V4: Experimental Evaluation

Editable manuscript section and supplementary results

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This draft accompanies the V4 methods package. It separates published reference scores, completed matched-data retraining, and retrospective Case1 recovery. All tables use saved results; no new model evaluations were performed to prepare this document.

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5 Experiments

We evaluate V4 on two complementary benchmark tasks and one retrospective case study: reaction-conditioned enzyme screening on EnzymeMap, bidirectional enzyme-reaction retrieval on ReactZyme, and literature-supported catalyst recovery for the conversion of D-fructose to D-tagatose. Following the task-based organization of FGW-CLIP [13], each benchmark is described in terms of its data, comparators, evaluation metrics, and results. We then analyze controlled changes to the training objective and inference settings using the recorded ablation experiments.

Common experimental design. V4 denotes one architecture and hyperparameter recipe fitted *separately* on each target training partition: the three ReactZyme splits and the EnzymeMap split. The trainable F3 retrieval layers are freshly initialized for every target; target-trained ReactZyme weights are not reused to fit EnzymeMap. Frozen protein, molecular, reaction, and SLEEC feature extractors remain pretrained resources. Phase 1 uses the anchor-balanced, decoupled multi-positive objective and the two attention regularizers described in the companion methods section. Phase 2 freezes F3 and fits the existing residual heads with full-graph uniform-positive cross-entropy and an identity-preservation penalty. The training-reaction dictionary is constructed from the target training graph only. Neither ensembling nor hubness correction is used.

The primary recipe retains phase-1 epoch 10 and phase-2 update 100 with seed 42. Its inference settings are dictionary weight $\alpha = 0.40$, protein fusion multiplier $\nu = 3$, and residual multiplier $c = 0.5$. The last parameter scales the residual update; it is not a norm-clipping threshold. V4+Bio adds relative biological supervision only to phase 2, with weight 1 for each of the EC, cofactor, and coarse transformation families, margin 0.1, and a fixed family divisor of three. This variant shares the original F3 checkpoints and adds no inference-time parameters. Detailed optimization settings appear in Appendix A.

Matching downstream training associations is necessary for our comparisons, but does not match every pretrained representation or annotation resource. We therefore distinguish comparisons between complete methods from controlled component ablations. The final recipe was reached through repeated exploratory benchmark inspection; fixing its final checkpoints does not make the overall development process a single-use held-out test. Reported differences are point estimates unless a paired-seed analysis is explicitly identified.

5.1 Enzyme screening on EnzymeMap

5.1.1 Datasets and screening protocols

We use the EnzymeMap release and reaction-rule split distributed with CLIPZyme [4, 6]. The processed dataset contains 46,356 enzyme-reaction association rows, 16,776 unique reactions, 12,749 enzymes, 2,841 EC identifiers, and 394 reaction rules. The released training, validation, and test partitions contain 34,427, 7,287, and 4,642 rows, respectively. The nominal 80/10/10 allocation is made over reaction rules; it does not imply these exact proportions of association rows. We retain the released partitions and physical reactant/product sides rather than resampling the split.

The screening library combines EnzymeMap, BRENDA release 2022_2, and UniProt release 2022_01, with a maximum sequence length of 650 residues. Its 261,907 candidate identifiers and their ordering are retained. Query construction follows the released CLIPZyme analysis notebook: the 4,642 test associations contain 1,551 distinct reaction strings, of which 30 also occur in training. Removing those reactions leaves 1,521 unique test queries. Known positive candidate identifiers are collected from the released raw EnzymeMap records and intersected with the screening library, yielding 4,544 query-candidate positive labels. These labels are used for scoring, not to extend the training association graph.

We report two screening settings. *Full pool* ranks all 261,907 candidates for each of the 1,521 queries. *Training-ID-excluded pool* removes every candidate identifier observed in the training associations, leaving 252,113 candidates and 1,337 queries with at least one remaining positive. This second setting follows the additional generalization analysis in FGW-CLIP. It tests exclusion of

training identifiers, not an independently established sequence-disjoint or homology-disjoint split. The existing ablation figures refer to these settings as Table 1 and Table 2, respectively; those labels are protocol names inherited from the source comparison, not manuscript table cross-references.

5.1.2 Baselines

We compare with the published CLIPZyme encoder variants and FGW-CLIP [6, 13]. We also evaluate the released CLIPZyme checkpoint on the same query lists, candidate order, positive labels, and screening metric implementation used for V4. Its downstream training data are the corresponding released EnzymeMap training partition; it is a checkpoint reproduction, not a new training run. Published FGW-CLIP numbers remain explicitly marked as reported values: no official checkpoint or verified reproducible implementation was available in the recorded source audit. Thus the CLIPZyme comparison has local score-level evidence, whereas the FGW-CLIP comparison is against its reported point estimates.

5.1.3 Evaluation metrics

We measure early retrieval with BEDROC at $a = 85$ and $a = 20$, and enrichment factors over the top 5% and 10% of each ranked library. Higher values are better. The screening tables show BEDROC as a percentage; the ablation figures and saved numerical files use its unscaled value. Enrichment factors are fold enrichments, not percentages. Metrics are calculated per reaction and averaged over eligible queries with equal query weight.

For a query with N candidates, n positives, and binary ranked labels $y_{(j)}$, the released enrichment calculation is

$$\text{EF}_f = \frac{\sum_{j=1}^{\lfloor fN \rfloor} y_{(j)}}{fn}, \quad f \in \{0.05, 0.10\}. \quad (1)$$

We preserve the notebook’s rank origin, sorting convention, and finite-library normalization for BEDROC; Appendix B gives the executable definition. MRR is not the screening endpoint or the criterion for interpreting these experiments.

5.1.4 Results

Tables 1 and 2 present all four screening metrics. In the full pool, V4 achieves BEDROC_{85} of 57.2337%, BEDROC_{20} of 75.5433%, EF5 of 16.8590, and EF10 of 8.9587. Its BEDROC_{85} exceeds the reproduced CLIPZyme value by 12.5394 percentage points and the reported FGW-CLIP value by 8.5737 percentage points. After excluding training enzyme identifiers, BEDROC_{85} remains 52.8734%, with EF5 of 16.4186 and EF10 of 8.7931. These results support improved early recovery under both recorded screening protocols.

Table 1: EnzymeMap screening against the full library: 1,521 queries and 261,907 candidate identifiers. BEDROC is expressed as a percentage. Published rows come from FGW-CLIP Table 1; locally measured rows use the shared released CLIPZyme notebook protocol. All V4 rows use the same target-specific training associations.

Method	BEDROC_{85} (%)	BEDROC_{20} (%)	EF5	EF10
CLIPZyme (ESM), reported	36.91	53.04	11.93	6.84
CLIPZyme (CGR), reported	38.91	57.58	13.16	7.73
CLIPZyme, reported	44.69	62.98	14.09	8.06
FGW-CLIP, reported	48.66	66.69	14.91	8.18
CLIPZyme, released checkpoint	44.6942	62.9817	14.0851	8.0600
V4	57.2337	75.5433	16.8590	8.9587
V4+Bio	57.2096	75.5322	16.8609	8.9565

Table 2: EnzymeMap screening after excluding training enzyme identifiers: 1,337 eligible queries and 252,113 candidate identifiers. Published values are from FGW-CLIP Table 2. The strongest published EF10 comparator in this setting is CLIPZyme (7.81), not FGW-CLIP (7.61).

Method	BEDROC ₈₅ (%)	BEDROC ₂₀ (%)	EF5	EF10
CLIPZyme, reported	39.13	58.86	13.40	7.81
FGW-CLIP, reported	45.14	61.43	13.57	7.61
CLIPZyme, released checkpoint	39.1328	58.8619	13.3967	7.8081
V4	52.8734	72.6403	16.4186	8.7931
V4+Bio	52.8464	72.6275	16.4357	8.8031

Adding biological supervision in phase 2 preserves the margins over those reported comparators, but does not improve all metrics over V4. BEDROC₈₅ decreases slightly in both pools; EF changes are small and mixed. We therefore attribute the screening result to the evaluated full recipe, without treating the biological objective as an established cause of the gain. The controlled label-removal and shuffle experiments below address that distinction directly.

5.2 Bidirectional enzyme–reaction retrieval on ReactZyme

5.2.1 Datasets and splits

ReactZyme links SwissProt proteins to Rhea reactions and provides temporal, enzyme-similarity, and reaction-similarity evaluation partitions [5]. The released resource is described as containing 178,463 association rows, 178,327 proteins, and 7,726 reaction identifiers. We use the existing materialized `reactzyme_paper` partitions shared by V4 and the public-baseline campaign. Their exact retained row counts appear in Table 3; these are the counts actually fitted and scored, rather than nominal totals copied from the paper. The materialized partitions contain 178,459 retained association rows per split. Published raw split sizes and retained distinct associations should not be interchanged.

Reaction similarity separates held-out reaction identifiers; enzyme similarity tests a sequence-based partition; and the temporal split separates annotations around the released 31 December 2010 cutoff. We preserve the supplied train/validation assignment, approximately a 90/10 subdivision of the non-test partition, and do not draw a new validation set for each comparator. The materialized reaction inputs are flattened participant sets. Their self-reaction encoding must not be interpreted as recovered physical reactant/product roles. EnzymeMap, in contrast, supplies those roles.

Table 3: Audited ReactZyme materialization. Train, validation, and test columns count retained association rows. The last two columns give the complete test reaction and enzyme catalogs: their roles as query and candidate sets swap between retrieval directions. Every locally retrained comparator uses these same catalogs and their recorded ordering.

Split	Train	Validation	Test	Reactions	Enzymes
Reaction similarity	147,393	16,377	14,689	386	14,688
Enzyme similarity	152,748	16,972	8,739	1,573	8,734
Time	149,554	16,618	12,287	2,634	12,277

5.2.2 Baselines and comparison scope

We separate two comparison tracks. The *published-reference track* includes FGW-CLIP’s EC Mode and EC Max variants and the two main TIGER variants, ESM2Text and ProtT3 [11, 13]. TIGER’s

main Reaction-Sim E→R MRR is 0.518 for ESM2Text; the 0.543 MLP ablation and 0.632 SwissProt-text condition are different experimental settings and are not relabeled as that main model.

The *matched-data retraining track* fits the four released ReactZyme families—MLP, contrastive MLP, Transformer, and Bi-RNN—with both ESM2 and SaProt protein features and four molecular feature choices (MAT-2D, MAT-3D, UniMol-2D, and UniMol-3D). This defines 96 family/feature/split configurations. An additional 24 configurations test a corrected contrastive-label convention and remain separate from the unchanged released contrastive model. We also include the completed original Horizyn participant-set adapter [7] and EnzGFM backbone controls. All completed configurations in the frozen evidence snapshot are retained in Appendix E; test performance is not used to select a representative feature combination.

The local baseline training uses the same downstream associations as V4, fresh retrieval weights, validation-only checkpoint selection, and a shared test evaluator. Material adaptations remain explicit: the released classifiers use reproducible sampled negatives because their original negative files are unavailable; SaProt follows the released ESM fallback for missing structural strings; Horizyn consumes a participant-set reaction adapter; and the cached EnzGFM preprocessing differs from its native paper convention. Their objectives, training budgets, and backbones are not identical. Consequently these are matched-data *method* comparisons, not isolated tests of architecture alone. Pending CREEP, structural CLIPZyme, EnzymeCAGE, and VenusRXN experiments have no completed result asserted here.

5.2.3 Evaluation metrics

For each direction, we rank the complete corresponding test candidate catalog. Let $P(q)$ be the positive candidates for query q , and $\text{rank}_q(p)$ their one-based ranks. Our primary metric is the released benchmark implementation’s all-positive, query-averaged MRR:

$$\text{MRR}_{\text{all}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|P(q)|} \sum_{p \in P(q)} \frac{1}{\text{rank}_q(p)}. \quad (2)$$

We additionally retain first-positive MRR and Hit@ k for $k = 1, 5, 10$ as diagnostics (Appendix B); they are not interchangeable with Eq. (2). Scores are ranked in FP32 with stable candidate-index tie breaking in the local shared evaluator.

Some published descriptions, including FGW-CLIP’s metric formula, instead describe first-positive MRR. Without its predictions or runnable evaluator, we cannot independently reconcile that wording with its reported tables. Published MRR columns are therefore reproduced *as reported* and kept separate from the common-evaluator retraining table. Local comparisons retain one metric definition and identical candidate axes throughout.

5.2.4 Results

Table 4 places V4 and V4+Bio alongside the published reference values. Original V4 records R→E/E→R MRR of 0.400381/0.534366 on Reaction-Sim, 0.666249/0.971355 on Enzyme-Sim, and 0.538285/0.780669 on Time. These exceed the displayed main-method literature point estimates, subject to the metric and resource qualifications above. The biology variant produces small R→E changes and larger, mixed E→R changes; numerical near ties qualify the latter, as examined in Section 5.4.4.

The newer matched-data results give a more restrictive conclusion (Table 5). The original Horizyn participant-set adapter obtains higher MRR than original V4 in all six split/direction cells. Its Reaction-Sim R→E MRR is 0.431815 versus 0.400381, and its Time R→E MRR is 0.598203 versus 0.538285. Horizyn also exceeds V4+Bio in all six cells, although the Reaction-Sim E→R margin is small (0.534799 versus 0.534777). Thus the completed local evidence does not support a claim that either V4 variant outperforms every retrained competitor. Different training budgets prevent assigning this difference to architecture alone. The full results are retained rather than excluding the stronger parent-model comparator.

Table 4: ReactZyme literature-reference comparison. The FGW-CLIP rows are transcribed from Tables 3, 7, and 8 of arXiv:2512.08508v2; TIGER rows are its main Table 1 values. Published MRR is retained as reported, while our rows use Eq. (2). This table is not a local reproduction of TIGER or FGW-CLIP, nor proof of equivalent pretraining resources or metric code.

Method	Reaction similarity		Enzyme similarity		Time	
	R→E	E→R	R→E	E→R	R→E	E→R
FGW-CLIP, EC Mode	0.3181	0.3722	0.5253	0.8683	0.3394	0.5222
FGW-CLIP, EC Max	0.3113	0.3804	0.5300	0.8581	0.3392	0.5229
TIGER (ESM2Text)	0.3190	0.5180	0.5920	0.9560	0.3660	0.6900
TIGER (ProtT3)	0.3370	0.4720	0.5790	0.9400	0.3720	0.6830
V4	0.400381	0.534366	0.666249	0.971355	0.538285	0.780669
V4+Bio	0.400412	0.534777	0.666234	0.967373	0.538317	0.800614

Table 5: Completed matched-data ReactZyme comparisons under the shared all-positive MRR evaluator. For the four released families and corrected-loss sensitivity, this compact table shows the same ESM2/MAT-2D features throughout; the EnzGFM control also uses MAT-2D. This is a fixed presentation slice, not the best feature pair chosen on test. All completed feature combinations are listed in Appendix E. Training budgets differ across methods.

Method	Reaction similarity		Enzyme similarity		Time	
	R→E	E→R	R→E	E→R	R→E	E→R
MLP	0.105690	0.218712	0.215499	0.707968	0.104607	0.284913
Contrastive MLP	0.040553	0.239766	0.119823	0.666660	0.040501	0.255020
Transformer	0.122008	0.231432	0.216317	0.725992	0.141887	0.394288
Bi-RNN	0.151047	0.283470	0.273579	0.796985	0.127391	0.375185
Corrected contrastive MLP	0.063832	0.243126	0.104661	0.631571	0.041036	0.241179
EnzGFM backbone control	0.135390	0.296464	0.286377	0.798907	0.152498	0.386741
Horizyn participant-set adapter	0.431815	0.534799	0.699701	0.976990	0.598203	0.840621
V4	0.400381	0.534366	0.666249	0.971355	0.538285	0.780669
V4+Bio	0.400412	0.534777	0.666234	0.967373	0.538317	0.800614

5.3 Case1: retrospective recovery in a small literature panel

5.3.1 Reaction and candidate construction

Case1 concerns the direct C4 epimerization of free D-fructose to D-tagatose, as investigated in the literature summarized by the source workbook [1–3, 8–10, 12, 14]. The workbook’s small homolog panel contains **145 entries**. One entry, H017 (KoT4E M6), lacks a fully specified recoverable sequence. The recorded evaluation therefore ranks **144 resolved entries**, representing **123 unique amino-acid sequences**. We report both the original-entry and deduplicated-sequence views; the missing sequence is not assigned an invented rank, and repeated entries are not counted as independent proteins.

We distinguish three evidence tiers. The conservative primary-paper set contains 12 unique sequences corresponding to 15 entries, with source-level activity evidence rechecked in the existing audit. Including patent evidence expands this set to 24 sequences and 36 entries. The broader workbook-active set contains 81 sequences and 102 entries; the remaining 42 unique sequences have only conditional non-detection evidence. Not every broader workbook assay has been independently verified, and non-detection under one assay condition is not a claim of universal inactivity. The workbook’s curated “Ranked Candidates” list is not used as binary experimental ground truth.

5.3.2 Deployment protocol and evaluation

The primary deployment uses the Reaction-Sim-trained checkpoint, with its recorded recipe frozen before that version’s Case1 scoring. No Case1 labels are used to fit the retrieval parameters. All other target-trained checkpoints are reported as transfer diagnostics without replacing the primary checkpoint on the basis of Case1 performance. Because the reaction and earlier outcomes were examined during development, this is retrospective evidence rather than an untouched external validation cohort.

For each candidate view, we report primary-paper recovery at ranks 5, 10, and 25 and normalize recall by that view’s own positive count. The unique view asks how many distinct documented sequences would be selected; the entry view preserves the requested workbook panel, including duplicate sequences and its deterministic tie order. As a descriptive reference, a uniform random ranking has expected primary-paper recovery at 25 of $25(12/123) = 2.439$ unique sequences or $25(15/144) = 2.604$ entries. These are expectations, not uncertainty estimates for the model.

We additionally report conditional AUROC for the 81 workbook-active versus 42 non-detect-only unique sequences. The same enzyme can appear in several constructs or studies, and assay conditions differ across sources. We therefore do not attach candidate-IID confidence intervals or interpret this AUROC as prospective activity-prediction accuracy.

5.3.3 Results and transfer limits

Table 6 and Figure 1 show the primary Reaction-Sim results. Original V4 recovers 10/12 unique paper-supported catalysts at rank 25, compared with 9/12 for its native F3 encoder before phase 2. Both recover 10/15 paper-supported entries in the 144-entry ranking. With phase-2 biological supervision, unique recovery increases to 11/12, whereas entry recovery remains 10/15. Conditional AUROC changes from 0.612875 to 0.613757. The unique-sequence gain is one catalyst, TsT4Ease WT (H008), moving from rank 26 to rank 25. Cofactor removal retains this gain; shuffled labels and mechanism removal do not.

Early cutoffs remain challenging: the primary final model recovers no primary-paper catalyst in its first five unique sequences and three in its first ten. Transfer also depends on the source training dataset: the EnzymeMap-trained V4 checkpoint recovers 6/12 at rank 25 with conditional AUROC 0.470606 (Appendix C). In the existing homology audit, four of the ten primary catalysts recovered by the Reaction-Sim model have retrieved training homologs with at least 90% identity; both query molecules also co-occur within larger training participant sets. Missing qualifying homology hits do not prove sequence novelty, and this audit does not cover frozen-backbone pretraining exposure.

Table 6: Case1 primary deployment and matched phase-2 controls. Unique recovery is out of 12 positives among 123 sequences; entry recovery is out of 15 positives among 144 resolved entries from the original 145-entry catalogue. AUROC is conditional on the broader 81/42 workbook labels. Controls use the same Reaction-Sim F3, phase-2 budget, and inference settings. “Native F3” is the saved pre-phase-2 diagnostic.

Model / labels	Unique @5	Unique @10	Unique @25	Entries @25	AUROC
Native F3	1	3	9	10	0.600823
V4	0	3	10	10	0.612875
V4+Bio	0	3	11	10	0.613757
Shuffled labels	0	3	10	10	0.611111
Without EC	0	3	10	10	0.613757
Without cofactor	0	3	11	10	0.614345
Without mechanism	0	3	10	10	0.613169

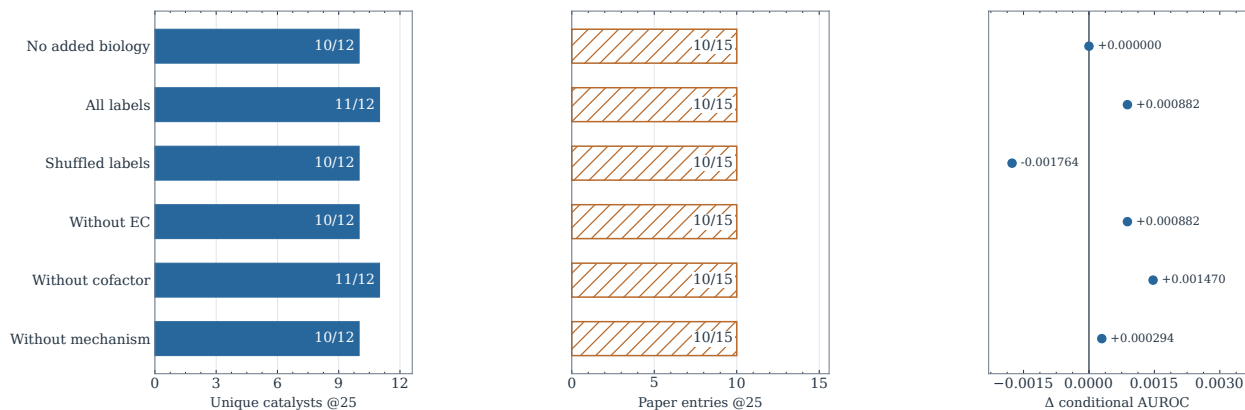


Figure 1: Case1 biological-supervision controls. Left: recovery at rank 25 among 123 unique sequences with 12 paper-supported catalysts. Middle: recovery among 144 resolved entries with 15 paper-supported entries. Right: conditional AUROC differences from original V4 (0.612875). All variants use the Reaction-Sim lineage and biological weight 1. The one-sequence gain in the deduplicated view does not improve the entry-level count. Colors, hatching, and printed counts distinguish the two evaluation units. These are retrospective literature results, not new wet-lab assays.

5.4 Ablation and sensitivity analyses

We retain the recorded controls that isolate a specific training or inference choice. Inference comparisons use identical trained weights and candidate pools. Phase-2 label controls hold F3 fixed. The replicated temperature and fresh-F3 biological studies use their own matched recipes and seeds; they are not presented as independent replications of the final single-seed V4 result. Appendix D lists the remaining component controls needed for a complete architectural ablation.

5.4.1 Dictionary weight and residual strength

At residual multiplier $c = 1$, increasing dictionary weight from 0.25 to 0.40 raises Reaction-Sim E→R MRR from 0.523804 to 0.537749. A further increase to 0.50 reduces R→E MRR on all three ReactZyme splits, while increasing EnzymeMap EF5 and EF10 in both pools. At $\alpha = 0.40$, reducing c from 1 to 0.5 lowers BEDROC₈₅ in both screening settings. These effects show a trade-off between retrieval directions and screening budgets, rather than a uniformly beneficial inference adjustment (Figures 2 and 3).

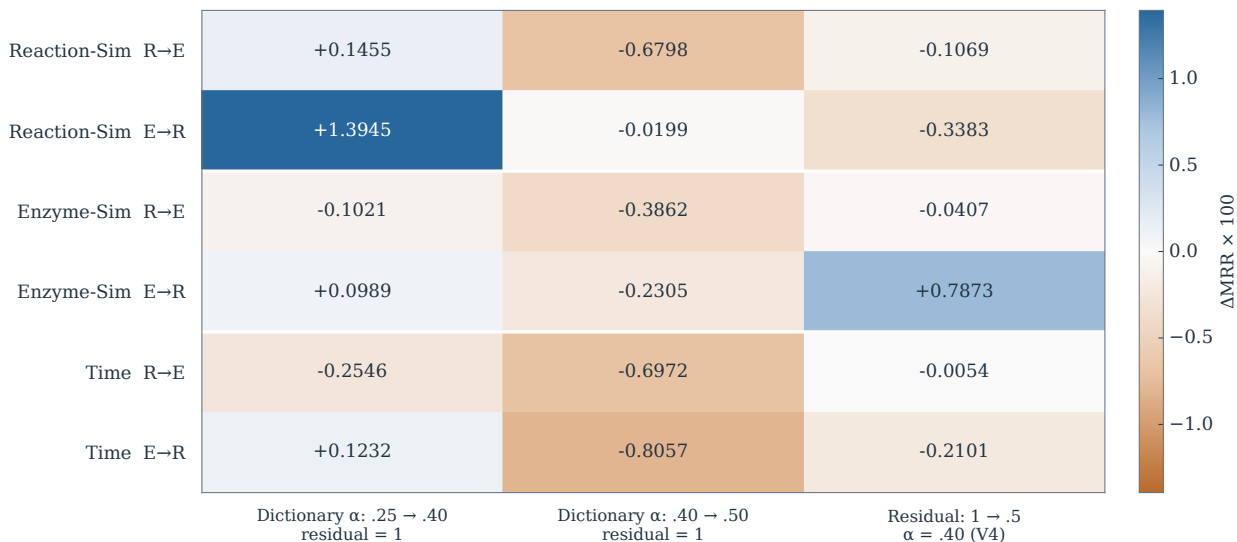


Figure 2: ReactZyme inference sensitivity with fixed checkpoints. Columns change one coefficient at a time; rows retain both directions and all three splits. Values are signed differences in MRR multiplied by 100; for example, +1.3945 represents +0.013945 raw MRR. The color scale is linear, centered at zero, with blue for positive and orange for negative changes. Larger Enzyme-Sim/Time E→R effects require the near-tie check in Figure 7.

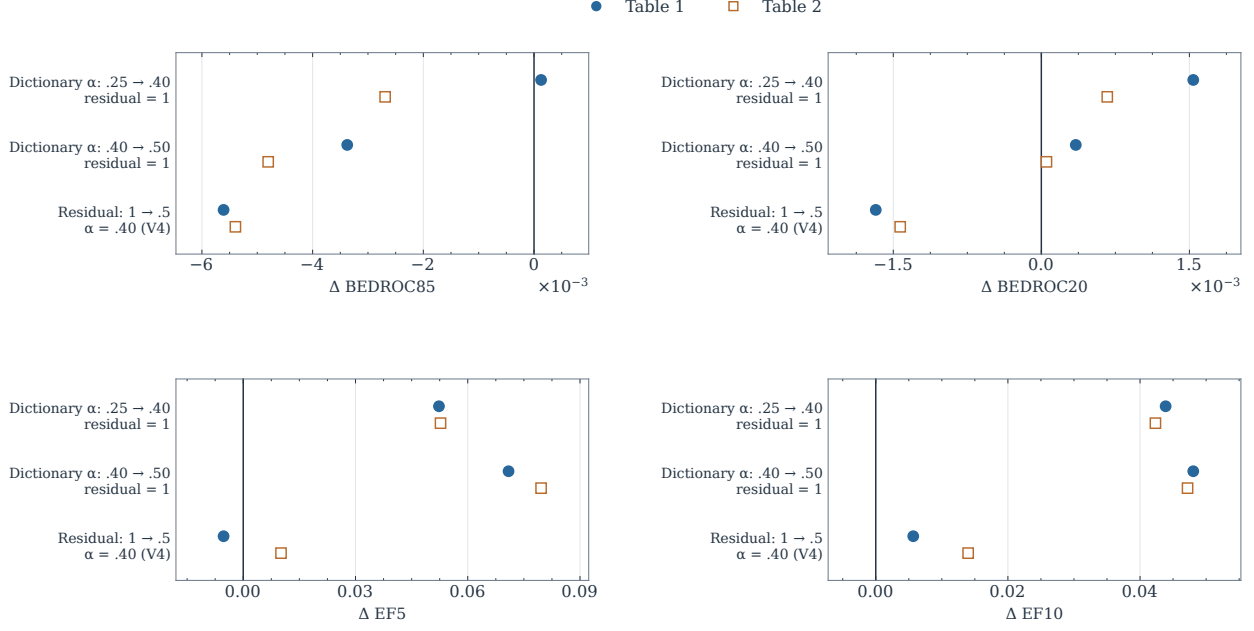


Figure 3: EnzymeMap inference sensitivity. Blue circles denote the full pool (Table 1 protocol); open orange squares denote training-ID exclusion (Table 2 protocol). Each point is the changed setting minus its stated reference; positive values favor the change. BEDROC differences are in raw units, not percentages. The four metric-specific axes prevent conflating enrichment with very-early recognition. Single-seed checkpoints and candidate pools are held fixed.

5.4.2 Anchor-balanced training and inverse temperature

The available replicated comparison isolates inverse temperature within an earlier 18-epoch EnzymeMap recipe: $\beta = 5$ versus $\beta = 10$, with paired seeds 17, 42, and 73 and the remaining training and phase-2 choices fixed. It does not isolate anchor balancing against the original MLNCE loss. Lower inverse temperature improves all BEDROC₈₅, BEDROC₂₀, and EF5 seed pairs. Overall, 22 of the 24 seed-by-metric comparisons improve; seed 73 produces the two negative EF10 differences. Full-pool mean BEDROC₈₅ increases from 0.494040 ± 0.035007 to 0.521656 ± 0.037549 , where the dispersion is sample standard deviation over three seeds. Figure 4 retains every seed rather than displaying only the mean. The metric cells are correlated, so their count is not a count of independent experiments.

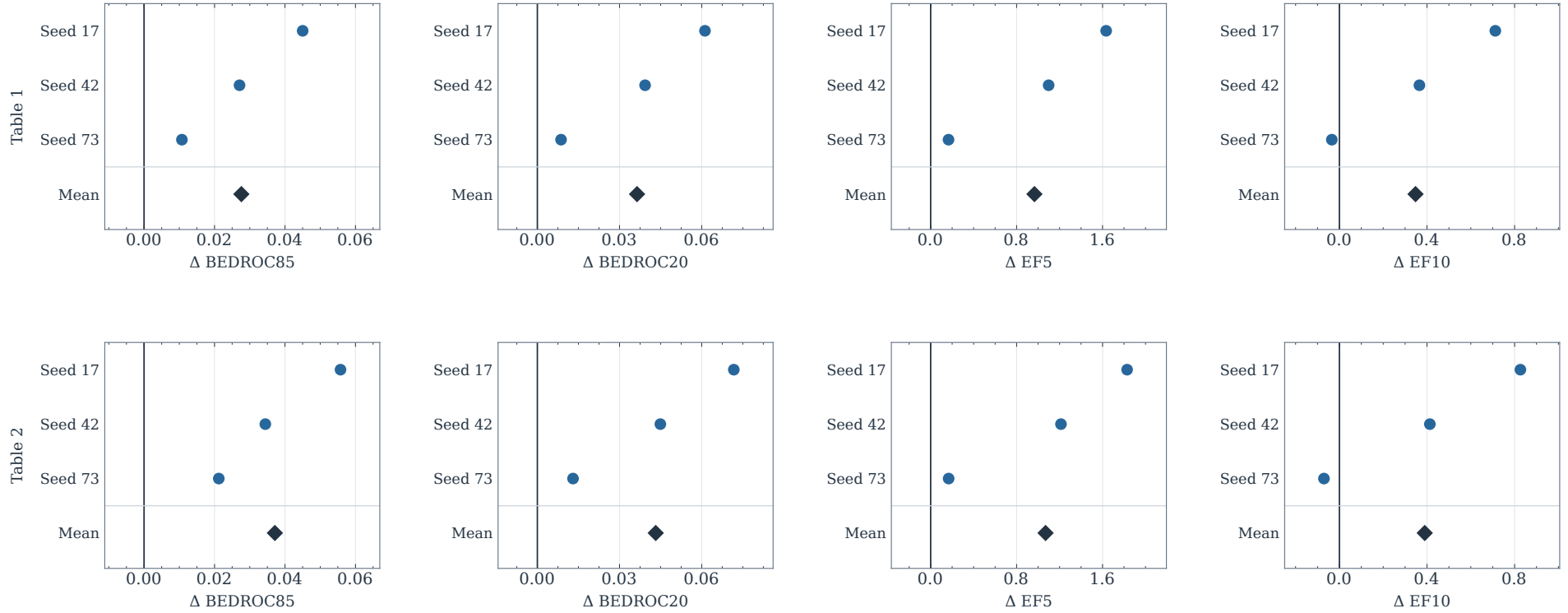


Figure 4: Paired inverse-temperature effects on EnzymeMap at fixed 18 epochs. Values are $\beta = 5$ minus $\beta = 10$; circles are individual seeds and dark diamonds their arithmetic mean. Upper row: full pool (Table 1). Lower row: training-ID-excluded pool (Table 2). Scales match across rows for each metric; BEDROC is unscaled. These observations describe a matched component study, not three-seed replication of the final 10-epoch V4 recipe or confidence intervals around that recipe.

5.4.3 Biological supervision in phase 2

We compare the unannotated V4 control with correctly assigned labels, shuffled labels, and removal of EC, cofactor, or transformation supervision. All active family coefficients are 1, and the averaging divisor remains three when a family is removed. Thus removal tests the family contribution at fixed remaining coefficients and also reduces total regularization; it is not an equal-total-weight reallocation.

The controls do not establish an overall benchmark advantage from adding biology. Full-pool BEDROC₈₅ is 0.572337 without added supervision, 0.572096 with correct labels, and 0.572093 with shuffled labels. Every displayed phase-2 variant slightly lowers both BEDROC metrics in both pools, while EF effects are mixed. ReactZyme point differences are likewise mixed (Figures 5 and 6). The Case1 controls in Figure 1 show a conditional top-25 crossing, not a consistent improvement across evaluation units and benchmarks.



Figure 5: ReactZyme phase-2 biological-supervision controls, each minus original V4. Cell values are $100\Delta\text{MRR}$. Colors use one shared symmetric logarithmic scale, linear within ± 0.001 displayed units ($\pm 10^{-5}$ raw MRR), so larger Time E→R differences do not wash out small effects elsewhere. Printed values are not log transformed; color intensity is nonlinear and is not directly comparable with the linear scale in Figure 2. Correct/shuffled labels use weight 1; removal controls retain other coefficients and the divisor of three. Visible color does not imply a practically large effect.

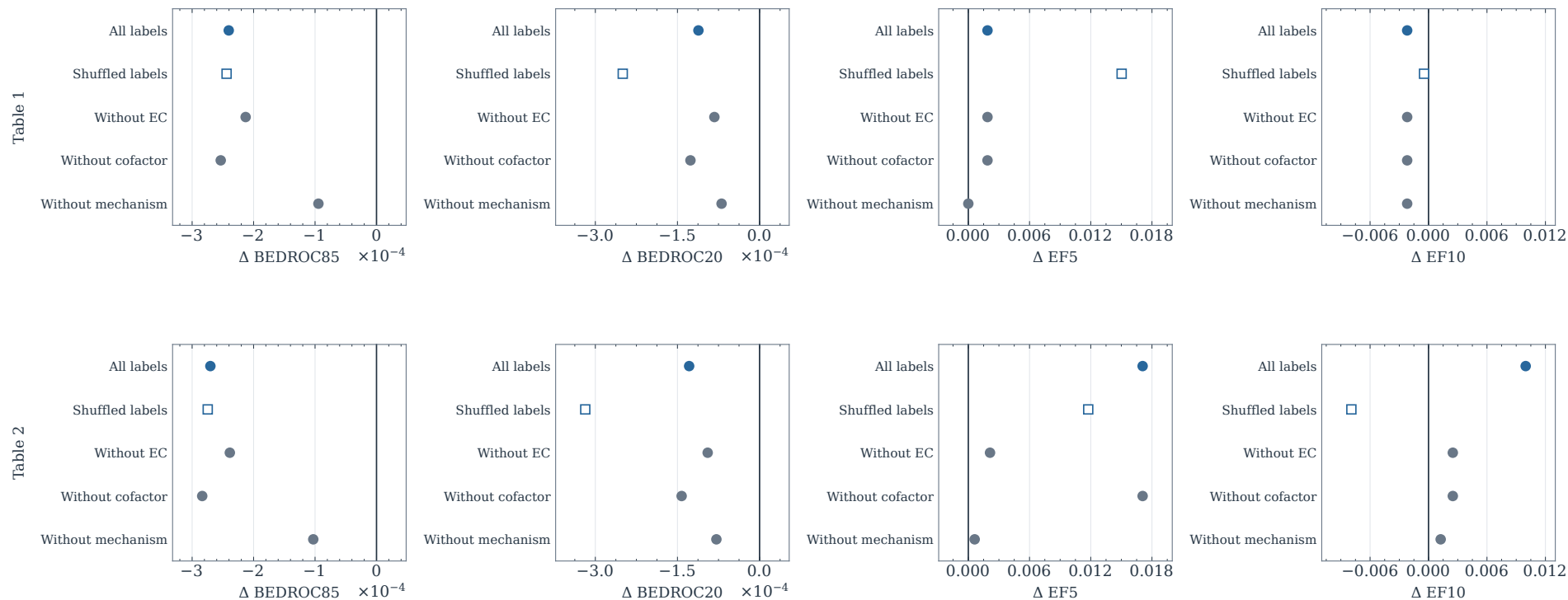


Figure 6: EnzymeMap phase-2 biological-supervision controls. Each variant is compared with original V4 without added supervision; F3, architecture, inference, pools, and training budget are fixed. Upper/lower rows are the full/training-ID-excluded pools. Filled blue circles denote correct labels, open squares shuffled labels, and gray circles family-removal controls. Metric scales match across rows. Axis multipliers matter: the BEDROC changes are on the order of 10^{-4} in raw units. Correct labels and shuffled labels nearly coincide on BEDROC_{85} .

5.4.4 Near-tie sensitivity of reaction retrieval

Some ReactZyme E→R metrics are sensitive to the ordering of nearly equal scores. We therefore retain the archived optimistic and pessimistic rank bounds obtained by permitting alternative orderings within a 10^{-6} score band. These are numerical sensitivity bounds, not confidence intervals. The small Reaction-Sim difference remains distinct in this diagnostic, whereas Enzyme-Sim and Time bounds overlap substantially (Figure 7). For example, the Time point estimate increases from 0.780669 to 0.800614 under phase-2 biology, but that increase alone does not establish robust improvement in reaction discrimination.

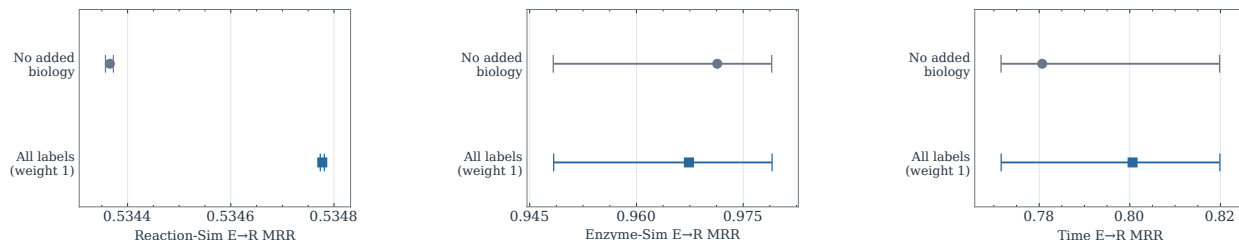


Figure 7: Recorded E→R MRR and numerical near-tie bounds for original V4 and phase-2 weight-1 biology. Points are recorded scores; line segments span optimistic and pessimistic ranks under a 10^{-6} score band. Each split has its own zoomed horizontal scale. Segment lengths should be compared within a panel, not across panels. The intervals quantify ranking sensitivity and have no sampling-uncertainty interpretation.

5.4.5 Biological supervision during F3 training

A separate EnzymeMap study retrains F3 with relative biological-loss weights 0, 0.1, or 1 across paired seeds 42, 43, and 44, keeping phase 2 unannotated. This directly changes F3 training, unlike V4+Bio. Weight 1 increases mean full-pool BEDROC₈₅ by 0.010256 but decreases mean EF5 by 0.151444. Only two of three seeds improve BEDROC₈₅; Table-2 EF5 decreases in all three seeds. Weight 0.1 improves Table-2 EF10 in all three seeds, but not all other metrics. Figure 8 exposes this seed and objective dependence. These results support a mixed biological-loss trade-off rather than a reproducible improvement on every early-screening endpoint.

5.5 Interpretation and remaining validation

The recorded experiments support improved EnzymeMap screening over the released CLIPZyme checkpoint and the displayed FGW-CLIP point estimates, and competitive ReactZyme scores relative to published main-method results. They also expose three boundaries: a newly retrained parent-model comparator is stronger on the six ReactZyme MRR cells; biological supervision has mixed or negligible controlled effects; and Case1 remains one dependent, retrospective literature panel. A claim of general superiority or broad real-world generalization would require the remaining matched controls, replication of the final recipe across seeds, and a prospectively specified external reaction panel. These unresolved questions are retained alongside the favorable benchmark results.

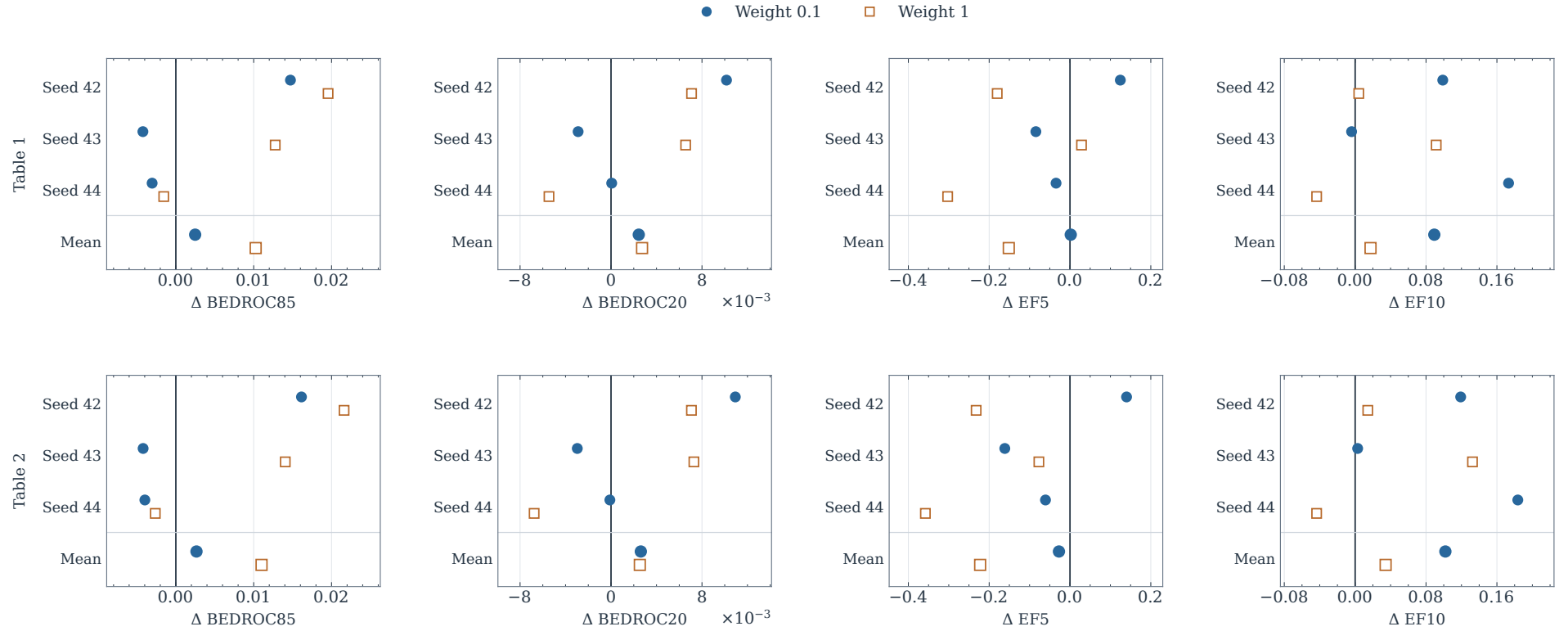


Figure 8: Fresh-F3 biological-loss study on EnzymeMap. Points are each biological variant minus its same-seed unannotated control; the last row in each panel is the arithmetic mean. Blue circles and open orange squares indicate weights 0.1 and 1. Upper/lower panels denote the full/training-ID-excluded pools, with matching metric scales. Phase 2 stays unannotated, and seed 42 uses the archived control. All seeds are displayed; these are not confidence intervals or estimates from the phase-2-only V4+Bio model.

A Implementation and comparison details

A.1 V4 optimization and inference

Table 7 summarizes the settings shared by the four target-specific fits. The pretrained feature extractors comprise ProtT5 residue representations, the frozen SLEEC scorer, ReactionT5v2, UniMol2, and ChIRo features. Their pretraining is not replaced by target-only training. The protein tower retains at most 1,022 residue vectors under the recorded ends/center policy; the EnzymeMap screening pool already restricts proteins to at most 650 residues. Feature extraction is reused across compatible experiments, while downstream retrieval layers and target training dictionaries remain target-specific.

Table 7: Fixed V4 recipe used for the primary benchmark tables. These settings are distinct from the earlier 18-epoch temperature study and the longer validation-selected public-baseline training runs.

Setting	Value
Seed and phase-1 batch size	42; 512 association rows
Phase-1 retained checkpoint	End of epoch 10
Phase-1 optimizer	AdamW; learning rate 10^{-4} ; weight decay 10^{-2}
Phase-1 retrieval objective	Anchor-balanced decoupled multi-positive loss; equal direction weights; fixed inverse temperature $\beta = 5$
Attention regularizers	Entropy and diversity coefficients both 0.01
Phase-2 retained checkpoint	Full-graph update 100; F3 frozen
Phase-2 optimizer	AdamW; learning rate 10^{-4} ; weight decay 10^{-3} ; gradient norm clipped at 1
Phase-2 retrieval objective	Uniform-positive cross-entropy; temperature 0.2
Identity penalty / training residual	10 / 0.2
Biological variant	Phase-2 relative loss; margin 0.1; coefficients (1, 1, 1); divisor 3
Inference multipliers	Protein fusion $\nu = 3$; residual $c = 0.5$
Training-reaction dictionary	Score weight $\alpha = 0.40$; 32 protein neighbors; protein and reaction kernel temperatures 0.03
Ensemble / hubness correction	Neither used

The dictionary and chemistry normalization are fitted from training inputs. Test associations are not added to the dictionary. Biological supervision uses annotations attached to training endpoints; the ReactZyme variant can also use reaction-only Rhea information. This is additional annotation supervision, not merely a different way of writing the same association loss. Conversely, the original V4 chemistry representation already contains cofactor-related features, so “without added biology” does not mean that every biological input has been removed.

The two active phases have different losses. Phase 1 normalizes positive contributions by anchor and uses a decoupled denominator. Phase 2 uses a uniform-positive cross-entropy over the complete training graph. The original global MLNCE loss is used by the separate parent Horizyn comparator; it is not the V4 phase-1 objective. Phase-2 biological labels cannot modify frozen residue queries, and their gradients alone do not establish biochemical interpretations for the learned views.

A.2 Public-baseline training adaptations

The four ReactZyme family implementations are imported from the released source. Their downstream classifiers use hidden dimension 128, output dimension 64, dropout 0, batch size 1,000, and AdamW with learning rate 10^{-4} and weight decay 5×10^{-10} . Training allows at most 200 epochs, with early stopping after 30 epochs without validation improvement; the retained checkpoint minimizes validation binary cross-entropy. Four deterministic corruptions per positive are used because the original negative pair files are absent. Known training positives are excluded from training negative samples; validation negatives also exclude validation positives. Positive and negative weights

are 1 and 0.25. This matches downstream association partitions, but is not an exact reproduction of published classifier scores.

The released contrastive implementation uses opposite label semantics in its contrastive and BCE branches. We retain that implementation under the “contrastive” label and report a separately named `contrastive_corrected` sensitivity that inverts the contrastive labels. We do not select between them using test results or relabel the correction as the unchanged published baseline.

The original Horizyn comparator preserves the public 2,048-to-4,096-to-4,096-to-512 reaction tower and 1,024-to-4,096-to-4,096-to-512 protein tower, global MLNCE with $\beta = 10$, normalized retrieval outputs, and original fingerprint generators. Its participant-set adapter writes the benchmark components on one side of the reaction representation. Training uses batch size 16,384, AdamW learning rate 10^{-4} , weight decay 0.01, and at most 100 epochs. Checkpoint selection maximizes mean bidirectional validation all-positive MRR evaluated every five epochs. The selected epochs are 100, 95, and 100 for Reaction-Sim, Enzyme-Sim, and Time. This is a materially larger training budget than V4’s ten retained epochs, so the observed ordering does not isolate the loss or architecture effect.

ESM2 and SaProt features follow the recorded released extraction conventions; 29,486 proteins without SaProt structural strings use the released ESM2 fallback. MAT and UniMol inputs preserve the supplied molecular participants and log conformer failures. The UniMol baseline uses UniMol v1, not the Uni-Mol2 features supplied to V4. EnzGFM rows use frozen 650M residue means with the cached 1,022-residue ends/center policy and special tokens excluded; the published extraction convention differs. These rows are therefore backbone controls. Reused frozen test-input embeddings do not imply reuse of target-trained V4 retrieval embeddings.

Completion snapshot. The captured evidence contains 126 completed public-model runs: 95 of the 96 unmodified official-family configurations, all 24 corrected-contrastive sensitivities, three Horizyn fits, and four EnzGFM controls. The remaining unmodified configuration is the Time-split Transformer with SaProt/UniMol-3D. Three CREEP runs and eight other EnzGFM configurations did not yet have completed test receipts in this snapshot. Full structural CLIPZyme and EnzymeCAGE comparisons require complete validated structures/pockets and justified reaction-input adapters; VenusRXN requires mapped directional inputs. Their missing results are not replaced with transferred checkpoints or reduced candidate pools. The supplementary tables are a frozen snapshot, not a live job-status page.

B Metric definitions and retrieval diagnostics

B.1 Released screening metric convention

For a ranked screening query, let $r_i \in \{0, \dots, N - 1\}$ be the *zero-based* indices of its n positive candidates and $\rho = n/N$. The released CLIPZyme notebook implementation reproduced in our evaluator is

$$S_a = \sum_{i=1}^n \exp(-ar_i/N), \quad (3)$$

$$D_a = \frac{\rho(1 - \exp(-a))}{\exp(a/N) - 1}, \quad (4)$$

$$F_a = \frac{\rho \sinh(a/2)}{\cosh(a/2) - \cosh(a/2 - a\rho)}, \quad (5)$$

$$\text{BEDROC}_a = \frac{S_a F_a}{D_a} + \frac{1}{1 - \exp(a(1 - \rho))}, \quad a \in \{85, 20\}. \quad (6)$$

This documents the implemented rank origin explicitly; we do not silently substitute a one-based formula or clip its finite-size output. The published CLIPZyme point estimates are closely reproduced under this convention. The enrichment cutoff is $\lfloor fN \rfloor$, while its denominator remains fn ,

as in Eq. (1). We preserve NumPy’s recorded descending score sorting convention for screening, independently of the stable candidate-index convention in the ReactZyme evaluator. Equal-score handling must not be silently unified across the two benchmark protocols.

B.2 All-positive and first-positive retrieval

In addition to Eq. (2), the saved ReactZyme scorer returns

$$\text{MRR}_{\text{first}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\min_{p \in P(q)} \text{rank}_q(p)}, \quad (7)$$

$$\text{Hit @}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1} \left[\min_{p \in P(q)} \text{rank}_q(p) \leq k \right]. \quad (8)$$

We call the latter Hit@ k to avoid importing conflicting Top- k /Top- k -N terminology from different papers. Duplicate positive edges are removed before scoring. Query-level normalization prevents a query with many known catalysts from contributing proportionally more weight. Queries with no positive in a specified diagnostic stratum are omitted from that stratum’s mean, while its candidate universe remains fixed.

Table 8 shows both reciprocal-rank definitions and the available hit rates for the same saved score matrices. Mean Rank, precision-normalized Top- k , and Hit@20 are not fabricated from these summaries; they would require additional rank-level aggregation.

Table 8: Diagnostic ReactZyme metrics from the saved primary test summaries. “MRR” is the all-positive metric; “First MRR” uses only the earliest positive. Values are fractions. No checkpoint is reselected using this table.

Model	Split	Direction	MRR	First MRR	Hit@1	Hit@5	Hit@10
V4	Reaction similarity	R→E	0.400381	0.594198	0.528497	0.655440	0.743523
V4	Reaction similarity	E→R	0.534366	0.534383	0.436274	0.647399	0.696623
V4	Enzyme similarity	R→E	0.666249	0.926589	0.890655	0.971392	0.984107
V4	Enzyme similarity	E→R	0.971355	0.971502	0.950195	0.994848	0.997138
V4	Time	R→E	0.538285	0.754115	0.689066	0.832954	0.877752
V4	Time	E→R	0.780669	0.780883	0.674106	0.902338	0.936874
V4+Bio	Reaction similarity	R→E	0.400412	0.594189	0.528497	0.655440	0.746114
V4+Bio	Reaction similarity	E→R	0.534777	0.534794	0.436887	0.647808	0.696963
V4+Bio	Enzyme similarity	R→E	0.666234	0.926588	0.890655	0.971392	0.984107
V4+Bio	Enzyme similarity	E→R	0.967373	0.967520	0.942295	0.994733	0.997138
V4+Bio	Time	R→E	0.538317	0.754041	0.688686	0.833333	0.878132
V4+Bio	Time	E→R	0.800614	0.800828	0.713692	0.902501	0.937118

C Case1 results across target-trained checkpoints

Table 9: Case1 transfer diagnostics for every target-trained model. Unique and entry counts are primary-paper recovery at rank 25, out of 12 and 15 respectively. AUROC is conditional on 81 workbook-active and 42 non-detect-only unique sequences. Reaction-Sim is the primary deployment; the other rows do not replace it after examining outcomes.

Training split	V4			V4+Bio		
	Unique	Entries	AUROC	Unique	Entries	AUROC
Reaction similarity	10	10	0.612875	11	10	0.613757
Enzyme similarity	5	8	0.584656	5	8	0.585538
Time	5	7	0.634333	5	7	0.636390
EnzymeMap	6	6	0.470606	6	6	0.470018

For the EnzymeMap-trained checkpoint, biological supervision leaves the unique-paper recovery at 6/12 and changes conditional AUROC from 0.470606 to 0.470018. Thus strong EnzymeMap benchmark performance is not sufficient to establish useful ranking for this particular literature reaction panel. The lower transfer performance is retained, rather than reporting only the best target-trained checkpoint.

The primary evidence tier comprises eight paper sources: the UxaE engineering study of Shin and colleagues; the binding-pocket work of Xia and colleagues; the thermostable discovery study of Chen and colleagues; the hyperthermostable enzyme characterization of Chen and colleagues; the expression/bioconversion study of Zhao and colleagues; the engineering study of Wang and colleagues; the coordinated engineering study of Zhu and colleagues; and the mining/characterization study of Guo and colleagues [1–3, 8–10, 12, 14]. The source audit records which evidence came from an abstract, indexed full-text section, or supporting-information description. It does not imply that every construct-level assay in the workbook was independently repeated or fully revalidated. Full sequence and source mappings accompany the local audit artifacts identified in the package’s source manifest.

D Scope of the existing ablation evidence

The missing matched controls include dictionary removal ($\alpha = 0$), residual removal ($c = 0$), both removals together, and a native fusion multiplier of one with the other final-recipe choices fixed. Loss, direction, and learned-query contributions require corresponding matched retraining. Historical best checkpoints cannot be combined into a causal component table. We report the existing figures as exploratory controlled comparisons, without inventing the missing runs or treating all variants as independent replications.

Table 10: What is controlled by the recorded ablations. A sensitivity analysis over nonzero coefficients is not a component-removal experiment.

Choice	Available evidence and limit
Dictionary weight	Same weights; $\alpha = 0.25, 0.40, 0.50$ at $c = 1$ on both benchmarks. Does not remove the dictionary.
Residual strength	Same weights; $c = 1$ versus $c = 0.5$ at $\alpha = 0.40$. Does not remove phase 2.
Phase-2 biology	No added labels, correct labels, shuffled labels, and three family-removal controls at fixed F3. Single seed; removal lowers total regularization.
Inverse temperature	Paired seeds 17/42/73 at 18 epochs on EnzymeMap. Not a three-seed final-V4 experiment or an MLNCE-versus-anchor-balanced comparison.
F3 biology	Paired seeds 42/43/44 with weights 0/0.1/1 and unannotated phase 2. Distinct from the phase-2-only variant.
Directional encoder	Historical pilots exist; no complete matched final-V4 directional versus invariant comparison on both benchmarks is established by this audit.
Learned residue queries	A fixed-budget comparison of four versus one versus no learned views remains necessary.
Architecture versus loss	The stronger parent Horizyn result changes inputs, capacity, objective, batch size, and budget. It is not an isolated loss ablation.

E Complete recorded public-baseline results

Tables below retain every completed test receipt at 2026-09-22 00:04:47 UTC. Abbreviations: **esm2** is ESM2-650M; **saprot** uses the recorded SaProt/ESM2 fallback; **enzgfm650** is the adapted frozen EnzGFM backbone. **mat_2d**, **mat_3d**, **unimol_2d**, and **unimol_3d** identify molecular feature sets. Missing feature fields in the Horizyn adapter row mean its own native fingerprint/ProtT5 inputs, not missing model inputs. Epochs are validation-selected. All scores below are all-positive MRR and use the same complete test catalogs as V4.

Table 11: Completed matched-data runs: Reaction similarity. Every recorded configuration is retained; rows are not selected by test performance.

Family / adapter	Protein	Reaction	Epoch	R→E	E→R
birnn	esm2	mat_2d	29	0.151047	0.283470
birnn	esm2	mat_3d	20	0.141364	0.320658
birnn	esm2	unimol_2d	23	0.143440	0.318308
birnn	esm2	unimol_3d	29	0.142052	0.313806
birnn	saprot	mat_2d	23	0.113894	0.293993
birnn	saprot	mat_3d	26	0.113946	0.233916
birnn	saprot	unimol_2d	26	0.125769	0.285104
birnn	saprot	unimol_3d	22	0.113728	0.278580
contrastive_corrected	esm2	mat_2d	26	0.063832	0.243126
contrastive_corrected	esm2	mat_3d	38	0.060943	0.224593
contrastive_corrected	esm2	unimol_2d	61	0.077039	0.245657
contrastive_corrected	esm2	unimol_3d	72	0.067378	0.242781
contrastive_corrected	saprot	mat_2d	25	0.058795	0.228902
contrastive_corrected	saprot	mat_3d	25	0.050967	0.207046
contrastive_corrected	saprot	unimol_2d	34	0.057993	0.237400
contrastive_corrected	saprot	unimol_3d	33	0.057324	0.237389
contrastive	esm2	mat_2d	73	0.040553	0.239766
contrastive	esm2	mat_3d	67	0.043296	0.238720
contrastive	esm2	unimol_2d	68	0.033340	0.178929
contrastive	esm2	unimol_3d	73	0.019032	0.179270
contrastive	saprot	mat_2d	85	0.033512	0.245238
contrastive	saprot	mat_3d	74	0.040848	0.226484
contrastive	saprot	unimol_2d	66	0.037882	0.315695
contrastive	saprot	unimol_3d	68	0.030569	0.260721
horizyn_participant_set	–	–	100	0.431815	0.534799

Family / adapter	Protein	Reaction	Epoch	R→E	E→R
mlp	esm2	mat_2d	61	0.105690	0.218712
mlp	esm2	mat_3d	92	0.109936	0.258306
mlp	esm2	unimol_2d	60	0.086496	0.214756
mlp	esm2	unimol_3d	68	0.100537	0.245804
mlp	saprot	mat_2d	96	0.102422	0.254348
mlp	saprot	mat_3d	43	0.101501	0.207088
mlp	saprot	unimol_2d	39	0.077519	0.290615
mlp	saprot	unimol_3d	104	0.115039	0.279126
transformer	enzgfm650	mat_2d	53	0.135390	0.296464
transformer	enzgfm650	mat_3d	24	0.107978	0.266878
transformer	esm2	mat_2d	26	0.122008	0.231432
transformer	esm2	mat_3d	48	0.146864	0.266343
transformer	esm2	unimol_2d	52	0.125993	0.270835
transformer	esm2	unimol_3d	28	0.100787	0.307673
transformer	saprot	mat_2d	47	0.112577	0.248236
transformer	saprot	mat_3d	33	0.117991	0.228077
transformer	saprot	unimol_2d	46	0.116645	0.263947
transformer	saprot	unimol_3d	42	0.112881	0.254783

Table 12: Completed matched-data runs: Enzyme similarity. Every recorded configuration is retained; rows are not selected by test performance.

Family / adapter	Protein	Reaction	Epoch	R→E	E→R
birnn	esm2	mat_2d	36	0.273579	0.796985
birnn	esm2	mat_3d	32	0.283009	0.800479
birnn	esm2	unimol_2d	26	0.279413	0.797344
birnn	esm2	unimol_3d	22	0.254641	0.789284
birnn	saprot	mat_2d	24	0.210042	0.733203
birnn	saprot	mat_3d	20	0.188961	0.710997
birnn	saprot	unimol_2d	26	0.212066	0.740293
birnn	saprot	unimol_3d	20	0.199273	0.728435
contrastive_corrected	esm2	mat_2d	29	0.104661	0.631571
contrastive_corrected	esm2	mat_3d	25	0.112843	0.602566
contrastive_corrected	esm2	unimol_2d	35	0.098874	0.615604
contrastive_corrected	esm2	unimol_3d	48	0.135219	0.705768
contrastive_corrected	saprot	mat_2d	30	0.067626	0.527562
contrastive_corrected	saprot	mat_3d	43	0.088303	0.593146
contrastive_corrected	saprot	unimol_2d	23	0.067232	0.539864
contrastive_corrected	saprot	unimol_3d	32	0.072214	0.569024
contrastive	esm2	mat_2d	67	0.119823	0.666660
contrastive	esm2	mat_3d	56	0.096680	0.620635
contrastive	esm2	unimol_2d	85	0.107420	0.650696
contrastive	esm2	unimol_3d	61	0.065233	0.536463
contrastive	saprot	mat_2d	56	0.075582	0.559432
contrastive	saprot	mat_3d	86	0.084863	0.647888
contrastive	saprot	unimol_2d	57	0.061696	0.538967
contrastive	saprot	unimol_3d	71	0.067325	0.586285
horizyn_participant_set	—	—	95	0.699701	0.976990
mlp	esm2	mat_2d	40	0.215499	0.707968
mlp	esm2	mat_3d	27	0.184238	0.678472
mlp	esm2	unimol_2d	92	0.250146	0.764819
mlp	esm2	unimol_3d	64	0.228901	0.760377
mlp	saprot	mat_2d	88	0.209825	0.737009
mlp	saprot	mat_3d	58	0.182154	0.682060
mlp	saprot	unimol_2d	55	0.195097	0.706072
mlp	saprot	unimol_3d	102	0.225738	0.736062
transformer	enzgfm650	mat_2d	40	0.286377	0.798907
transformer	esm2	mat_2d	22	0.216317	0.725992
transformer	esm2	mat_3d	28	0.246319	0.770046
transformer	esm2	unimol_2d	27	0.231870	0.777284

Family / adapter	Protein	Reaction	Epoch	R→E	E→R
transformer	esm2	unimol_3d	49	0.280637	0.807310
transformer	saprot	mat_2d	45	0.211748	0.738392
transformer	saprot	mat_3d	45	0.204797	0.720958
transformer	saprot	unimol_2d	41	0.229146	0.753961
transformer	saprot	unimol_3d	24	0.171613	0.701101

Table 13: Completed matched-data runs: Time. Every recorded configuration is retained; rows are not selected by test performance.

Family / adapter	Protein	Reaction	Epoch	R→E	E→R
birnn	esm2	mat_2d	28	0.127391	0.375185
birnn	esm2	mat_3d	18	0.125936	0.351964
birnn	esm2	unimol_2d	27	0.143554	0.405404
birnn	esm2	unimol_3d	22	0.134341	0.357911
birnn	saprot	mat_2d	45	0.057523	0.147505
birnn	saprot	mat_3d	22	0.053898	0.127134
birnn	saprot	unimol_2d	19	0.042792	0.114786
birnn	saprot	unimol_3d	31	0.056384	0.131066
contrastive_corrected	esm2	mat_2d	22	0.041036	0.241179
contrastive_corrected	esm2	mat_3d	27	0.038138	0.240529
contrastive_corrected	esm2	unimol_2d	20	0.038952	0.231652
contrastive_corrected	esm2	unimol_3d	27	0.035515	0.241420
contrastive_corrected	saprot	mat_2d	50	0.020605	0.108698
contrastive_corrected	saprot	mat_3d	54	0.019258	0.092859
contrastive_corrected	saprot	unimol_2d	41	0.024489	0.088000
contrastive_corrected	saprot	unimol_3d	46	0.025426	0.096020
contrastive	esm2	mat_2d	82	0.040501	0.255020
contrastive	esm2	mat_3d	72	0.036686	0.275647
contrastive	esm2	unimol_2d	104	0.028089	0.274003
contrastive	esm2	unimol_3d	61	0.017047	0.215914
contrastive	saprot	mat_2d	61	0.015760	0.093115
contrastive	saprot	mat_3d	57	0.014666	0.096924
contrastive	saprot	unimol_2d	59	0.013870	0.101722
contrastive	saprot	unimol_3d	53	0.011945	0.086796
horizyn_participant_set	–	–	100	0.598203	0.840621
mlp	esm2	mat_2d	73	0.104607	0.284913
mlp	esm2	mat_3d	75	0.109812	0.296219
mlp	esm2	unimol_2d	74	0.103276	0.299622
mlp	esm2	unimol_3d	55	0.108378	0.288156
mlp	saprot	mat_2d	101	0.056410	0.141454
mlp	saprot	mat_3d	33	0.040105	0.093413
mlp	saprot	unimol_2d	46	0.047138	0.104551
mlp	saprot	unimol_3d	55	0.043459	0.098773
transformer	enzgfm650	mat_2d	57	0.152498	0.386741
transformer	esm2	mat_2d	64	0.141887	0.394288
transformer	esm2	mat_3d	29	0.137216	0.355277
transformer	esm2	unimol_2d	45	0.148942	0.398539
transformer	esm2	unimol_3d	29	0.122346	0.357867
transformer	saprot	mat_2d	38	0.047058	0.111095
transformer	saprot	mat_3d	26	0.049423	0.130129
transformer	saprot	unimol_2d	59	0.055180	0.139714

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